

11.24.8.7 (ID 3551) Vibration Min Factor

Location in the menu: **P3.24.4.2.2**

Use this parameter to set the min factor in the threshold value. If the threshold value is zero, the min factor is not used.

11.24.8.8 (ID 3552) Vibration Max Factor

Location in the menu: **P3.24.4.2.3**

Use this parameter to set the max factor in the threshold value. If the threshold value is zero, the max factor is not used.

11.24.8.9 (ID 3553) Vibration Std Factor

Location in the menu: **P3.24.4.2.4**

Use this parameter to set the std factor in the threshold value. If the threshold value is zero, the std factor is not used.

11.24.8.10 (ID 3554) Vibration Interpolation Type

Location in the menu: **P3.24.4.2.5**

Use this parameter to select interpolation type for the notification threshold values across frequency points.

11.24.8.11 (ID 3567) Load Mean Factor

Location in the menu: **P3.24.5.1.1**

Use this parameter to set the mean factor in the threshold value. If the threshold value is zero, the mean factor is not used.

11.24.8.12 (ID 3568) Load Min Factor

Location in the menu: **P3.24.5.1.2**

Use this parameter to set the min factor in the threshold value. If the threshold value is zero, the min factor is not used.

11.24.8.13 (ID 3569) Load Max Factor

Location in the menu: **P3.24.5.1.3**

Use this parameter to set the max factor in the threshold value. If the threshold value is zero, the max factor is not used.

11.24.8.14 (ID 3570) Load Std Factor

Location in the menu: **P3.24.5.1.4**

Use this parameter to set the std factor in the threshold value. If the threshold value is zero, the std factor is not used.

11.24.8.15 (ID 3571) Load Interpolation Type

Location in the menu: **P3.24.5.1.5**

Use this parameter to select interpolation type for the notification threshold values across frequency points.

11.24.9 Warning and Alarm/Fault Stages

Each of the three detection levels (Warning S1, Warning S2, Alarm/Fault) has a configurable timer associated that delays the response. The timing principle is like an electrical fuse where the response time is long for low levels and it is short for high levels.

11.24.9.1 (ID 3516) Current Warning S1 Mode

Location in the menu: **P3.24.3.3.2.1**

Use this parameter to select the warning S1 mode which is used for calculating warning S1 notification threshold.

Selection number	Selection name	Description
0	Absolute	Absolute value is considered as threshold
1	Offset	Calculates threshold as sum of the computed baseline data and the offset values
2	Factor	Calculates threshold as baseline data factor

11.24.9.2 (ID 3517) Current Warning S1 High

Location in the menu: **P3.24.3.3.2.2**

Use this parameter to set the threshold value for computing the warning 1 high notification threshold. Threshold value range varies based on warning mode selection. This limit is not used when value is zero.

11.24.9.3 (ID 3518) Current Warning S1 Delay

Location in the menu: **P3.24.3.3.2.3**

Use this parameter to define how long time the actual value must be above the warning S1 notification level to trigger the warning S1. This level is not used when delay is zero.

11.24.9.4 (ID 3519) Current Warning S2 Mode

Location in the menu: **P3.24.3.3.2.4**

Use this parameter to select the warning S2 mode which is used for calculating warning S2 notification threshold.

Selection number	Selection name	Description
0	Absolute	Absolute value is considered as threshold
1	Offset	Calculates threshold as sum of the computed baseline data and the offset values
2	Factor	Calculates threshold as baseline data factor

11.24.9.5 (ID 3520) Current Warning S2 High

Location in the menu: **P3.24.3.3.2.5**

Use this parameter to set the threshold value for computing the warning 2 high notification threshold. Threshold value range varies based on warning mode selection. This limit is not used when value is zero.

11.24.9.6 (ID 3521) Current Warning S2 Delay

Location in the menu: **P3.24.3.3.2.6**

Use this parameter to define how long time the actual value must be above the warning S2 notification level to trigger the warning S2. This level is not used when delay is zero.

11.24.9.7 (ID 3522) Current Alarm/Fault Mode

Location in the menu: **P3.24.3.3.2.7**

Use this parameter to select the alarm/fault mode which is used for calculating alarm/fault notification threshold.

Selection number	Selection name	Description
0	Absolute	Absolute value is considered as threshold
1	Offset	Calculates threshold as sum of the computed baseline data and the offset values
2	Factor	Calculates threshold as baseline data factor

11.24.9.8 (ID 3523) Current Alarm/Fault High

Location in the menu: **P3.24.3.3.2.8**

Use this parameter to set the threshold value for computing the alarm/fault high notification threshold. Threshold value range varies based on alarm/fault mode selection. This limit is not used when value is zero.

11.24.9.9 (ID 3524) Current Alarm/Fault Delay

Location in the menu: **P3.24.3.3.2.9**

Use this parameter to define how long time the actual value must be above the alarm/fault notification level to trigger the alarm/fault. This level is not used when delay is zero.

11.24.9.10 (ID 3541) Current Warning S1 Counter

Location in the menu: **P3.24.3.3.1**

Use this parameter to set warning S1 counter value.

11.24.9.11 (ID 3542) Current Warning S2 Counter

Location in the menu: **P3.24.3.3.2**

Use this parameter to set warning S2 counter value.

11.24.9.12 (ID 3543) Current Alarm/Fault Counter

Location in the menu: **P3.24.3.3.3**

Use this parameter to set alarm/fault counter value.

11.24.9.13 (ID 3549) Stop Counter Delay

Location in the menu: **P3.24.2.3.4, P3.24.4.4.4, and P3.24.5.3.4**

Location in the menu: **P3.23.2.3.4, P3.23.4.4.4, and P3.23.5.3.4**

Use this parameter to set stop counter. The counter is stopped if threshold value is outside of the limit below this time.

11.24.9.14 (ID 3555) Vibration Warning S1 Mode

Location in the menu: **P3.24.4.3.1**

Use this parameter to select the warning S1 mode which is used for calculating warning S1 notification threshold.

Selection number	Selection name	Description
0	Absolute	Absolute value is considered as threshold
1	Offset	Calculates threshold as sum of the computed baseline data and the offset values
2	Factor	Calculates threshold as baseline data factor

11.24.9.15 (ID 3556) Vibration Warning S1 High

Location in the menu: **P3.24.4.3.2**

Use this parameter to set the threshold value for computing the warning 1 high notification threshold. Threshold value range varies based on warning mode selection. This limit is not used when value is zero.

11.24.9.16 (ID 3557) Vibration Warning S1 Delay

Location in the menu: **P3.24.4.3.3**

Use this parameter to define how long time the actual value must be above the warning S1 notification level to trigger the warning S1. This level is not used when delay is zero.

11.24.9.17 (ID 3558) Vibration Warning S2 Mode

Location in the menu: **P3.24.4.3.4**

Use this parameter to select the warning S2 mode which is used for calculating warning S2 notification threshold.

Selection number	Selection name	Description
0	Absolute	Absolute value is considered as threshold
1	Offset	Calculates threshold as sum of the computed baseline data and the offset values
2	Factor	Calculates threshold as baseline data factor

11.24.9.18 (ID 3559) Vibration Warning S2 High

Location in the menu: **P3.24.4.3.5**

Use this parameter to set the threshold value for computing the warning 2 high notification threshold. Threshold value range varies based on warning mode selection. This limit is not used when value is zero.

11.24.9.19 (ID 3560) Vibration Warning S2 Delay

Location in the menu: **P3.24.4.3.6**

Use this parameter to define how long time the actual value must be above the warning S2 notification level to trigger the warning S2. This level is not used when delay is zero.

11.24.9.20 (ID 3561) Vibration Alarm/Fault Mode

Location in the menu: **P3.24.4.3.7**

Use this parameter to select the alarm/fault mode which is used for calculating alarm/fault notification threshold.

Selection number	Selection name	Description
0	Absolute	Absolute value is considered as threshold
1	Offset	Calculates threshold as sum of the computed baseline data and the offset values
2	Factor	Calculates threshold as baseline data factor

11.24.9.21 (ID 3562) Vibration Alarm/Fault High

Location in the menu: **P3.24.4.3.8**

Use this parameter to set the threshold value for computing the alarm/fault high notification threshold. Threshold value range varies based on warning mode selection. This limit is not used when value is zero.

11.24.9.22 (ID 3563) Vibration Alarm/Fault Delay

Location in the menu: **P3.24.4.3.9**

Use this parameter to define how long time the actual value must be above the alarm/fault notification level to trigger the alarm/fault. This level is not used when delay is zero.

11.24.9.23 (ID 3564) Vibration Warning S1 Counter

Location in the menu: **P3.24.4.4.1**

Use this parameter to set warning S1 counter value.

11.24.9.24 (ID 3565) Vibration Warning S2 Counter

Location in the menu: **P3.24.4.4.2**

Use this parameter to set warning S2 counter value.

11.24.9.25 (ID 3566) Vibration Alarm/Fault Counter

Location in the menu: **P3.24.4.4.3**

Use this parameter to set alarm/fault counter value.

11.24.9.26 (ID 3572) Load Warning S1 Mode

Location in the menu: **P3.24.5.2.1**

Use this parameter to select the warning S1 mode which is used for calculating warning S1 notification threshold.

Selection number	Selection name	Description
0	Absolute	Absolute value is considered as threshold
1	Offset	Calculates threshold as sum of the computed baseline data and the offset values
2	Factor	Calculates threshold as baseline data factor

11.24.9.27 (ID 3573) Load Warning S1 High

Location in the menu: **P3.24.5.2.3**

Use this parameter to set the threshold value for computing the warning 1 high notification threshold. Threshold value range varies based on warning mode selection. This limit is not used when value is zero.

11.24.9.28 (ID 3574) Load Warning S1 Delay

Location in the menu: **P3.24.5.2.4**

Use this parameter to define how long time the actual value must be above the warning S1 notification level to trigger the warning S1. This level is not used when delay is zero.

11.24.9.29 (ID 3575) Load Warning S2 Mode

Location in the menu: **P3.24.5.2.5**

Use this parameter to select the warning S2 mode which is used for calculating warning S2 notification threshold.

Selection number	Selection name	Description
0	Absolute	Absolute value is considered as threshold
1	Offset	Calculates threshold as sum of the computed baseline data and the offset values
2	Factor	Calculates threshold as baseline data factor

11.24.9.30 (ID 3576) Load Warning S2 High

Location in the menu: **P3.24.5.2.7**

Use this parameter to set the threshold value for computing the warning 2 high notification threshold. Threshold value range varies based on warning mode selection. This limit is not used when value is zero.

11.24.9.31 (ID 3577) Load Warning S2 Delay

Location in the menu: **P3.24.5.2.8**

Use this parameter to define how long time the actual value must be above the warning S2 notification level to trigger the warning S2. This level is not used when delay is zero.

11.24.9.32 (ID 3578) Load Alarm/Fault Mode

Location in the menu: **P3.24.5.2.9**

Use this parameter to select the alarm/fault mode which is used for calculating alarm/fault notification threshold.

Selection number	Selection name	Description
0	Absolute	Absolute value is considered as threshold
1	Offset	Calculates threshold as sum of the computed baseline data and the offset values
2	Factor	Calculates threshold as baseline data factor

11.24.9.33 (ID 3579) Load Alarm/Fault High

Location in the menu: **P3.24.5.2.11**

Use this parameter to set the threshold value for computing the alarm/fault high notification threshold. Threshold value range varies based on warning mode selection. This limit is not used when value is zero.

11.24.9.34 (ID 3580) Load Alarm/Fault Delay

Location in the menu: **P3.24.5.2.12**

Use this parameter to define how long time the actual value must be above the alarm/fault notification level to trigger the alarm/fault. This level is not used when delay is zero.

11.24.9.35 (ID 3581) Load Warning S1 Counter

Location in the menu: **P3.24.5.3.1**

Use this parameter to set warning S1 counter value.

11.24.9.36 (ID 3582) Load Warning S2 Counter

Location in the menu: **P3.24.5.3.2**

Use this parameter to set warning S2 counter value.

11.24.9.37 (ID 3583) Load Alarm/Fault Counter

Location in the menu: **P3.24.5.3.3**

Use this parameter to set alarm/fault counter value.

11.24.9.38 (ID 3584) Load Warning S1 Low

Location in the menu: **P3.24.5.2.2**

Use this parameter to set the threshold value for computing the warning 1 low notification threshold. Threshold value range varies based on warning mode selection. This limit is not used when value is zero.

11.24.9.39 (ID 3585) Load Warning S2 Low

Location in the menu: **P3.24.5.2.6**

Use this parameter to set the threshold value for computing the warning 2 low notification threshold. Threshold value range varies based on warning mode selection. This limit is not used when value is zero.

11.24.9.40 (ID 3586) Load Alarm/Fault Low

Location in the menu: **P3.24.5.2.10**

Use this parameter to set the threshold value for computing the alarm/fault low notification threshold. Threshold value range varies based on warning mode selection. This limit is not used when value is zero.

11.24.10 (ID 3587) Vibration Input

Location in the menu: **P3.24.4.1**

Use this parameter to select the analog input to vibration.

11.25 Default Values of Parameters in the Different Applications

Table 132: Default Values of Parameters of Group 3.2

Index	Parameter	Standard application	HVAC application	PID control application	Multi-pump (single drive) application	Multi-pump (multidrive) application	Unit	ID	Description
3.2.1	Rem Control Place	0	0	0	0	0		172	0 = I/O Control
3.2.2	Local/Remote	0	0	0	0	0		211	0 = Remote
3.2.6	I/O A Logic	2	2	2	0	0		300	2 = Forw-Back (edge)
3.2.7	I/O B Logic	2	2	2	2	2		363	2 = Forw-Back (edge)

Table 133: Default Values of Parameters of Group 3.3

Index	Parameter	Standard application	HVAC application	PID control application	Multi-pump (single drive) application	Multi-pump (multi-drive) application	Unit	ID	Description
3.3.1.5	I/O A Ref Sel	6	6	7	7	7		117	6 = AI1 + AI2 7 = PID
3.3.1.6	I/O B Ref Sel	4	4	4	4	4		131	4 = AI1
3.3.1.7	Keypad Ref Sel	2	2	2	2	2		121	2 = Keypad Reference
3.3.1.10	Fieldbus Ref Sel	3	3	3	3	3		122	3 = Fieldbus Reference
3.3.3.1	Preset Freq Mode	0	0	0	0	0		182	0 = Binary Coded
3.3.3.3	Preset Freq 1	10.0	10.0	10.0	10.0	10.0	Hz	105	
3.3.3.4	Preset Freq 2	15.0	15.0	15.0	15.0	15.0	Hz	106	
3.3.3.5	Preset Freq 3	20.0	20.0	20.0	20.0	20.0	Hz	126	
P3.3.6.1	Flushing Reference Activation	0	0	0	0	101		530	0 = DigIN Slot0.1 101 = DigIN SlotA.2
P3.3.6.2	Flushing Reference	0	0	0	0	0	Hz	1239	

Table 134: Default Values of Parameters of Group 3.5

Index	Parameter	Standard application	HVAC application	PID control application	Multi-pump (single drive) application	Multi-pump (multi-drive) application	Unit	ID	Description
3.5.1.1	Ctrl Signal 1 A	100	100	100	100	100		403	100 = DigIN SlotA.1
3.5.1.2	Ctrl Signal 2 A	101	101	0	0	0		404	0 = DigIN Slot0.1 101 = DigIN SlotA.2
3.5.1.4	Ctrl Signal 1 B	0	0	103	101	0		423	0 = DigIN Slot0.1 101 = DigIN SlotA.2 103 = DigIN SlotA.4
3.5.1.7	I/O B Ctrl Force	0	0	105	102	0		425	0 = DigIN Slot0.1 102 = DigIN SlotA.3 105 = DigIN SlotA.6
3.5.1.8	I/O B Ref Force	0	0	105	102	0		343	0 = DigIN Slot0.1 102 = DigIN SlotA.3 105 = DigIN SlotA.6

Index	Parameter	Standard application	HVAC application	PID control application	Multi-pump (single drive) application	Multi-pump (multi-drive) application	Unit	ID	Description
3.5.1.9	Fieldbus Ctrl Force	0	0	0	0	0		411	
3.5.1.10	Keypad Ctrl Force	0	0	0	0	0		410	
3.5.1.11	External Fault Close	102	102	101	0	105		405	0 = DigIN Slot0.1 101 = DigIN SlotA.2 102 = DigIN SlotA.3 105 = DigIN SlotA.6
3.5.1.13	Fault Reset Close	105	105	102	0	103		414	0 = DigIN Slot0.1 102 = DigIN SlotA.3 103 = DigIN SlotA.4 105 = DigIN SlotA.6
3.5.1.21	Preset Freq Sel0	103	103	104	0	0		419	0 = DigIN Slot0.1 103 = DigIN SlotA.4 104 = DigIN SlotA.5
3.5.1.22	Preset Freq Sel1	104	104	0	0	0		420	0 = DigIN Slot0.1 104 = DigIN SlotA.5
3.5.1.23	Preset Freq Sel2	0	0	0	0	0		421	
P3.5.1.31	PID Setpoint Selection	0	0	0	0	102		1047	0 = DigIN Slot0.1 102 = DigIN SlotA.3
P3.5.1.36	Flushing Reference Activation	0	0	0	0	101		530	0 = DigIN Slot0.1 101 = DigIN SlotA.2
P3.5.1.42	Pump 1 Interlock	0	0	0	103	0		426	0 = DigIN Slot0.1 103 = DigIN SlotA.4
P3.5.1.43	Pump 2 Interlock	0	0	0	104	0		427	0 = DigIN Slot0.1 104 = DigIN SlotA.5
P3.5.1.44	Pump 3 Interlock	0	0	0	105	0		428	0 = DigIN Slot0.1 105 = DigIN SlotA.6
3.5.2.1.1	AI1 Signal Selection	100	100	100	100	100		377	100 = AnIN SlotA.1
3.5.2.1.2	AI1 Filter Time	0.1	0.1	0.1	0.1	0.1	s	378	
3.5.2.1.3	AI1 Signal Range	0	0	0	0	0		379	0 = 0...10 V/0...20 mA
3.5.2.1.4	AI1 Custom Min	0.0	0.0	0.0	0.0	0.0	%	380	
3.5.2.1.5	AI1 Custom Max	100.0	100.0	100.0	100.0	100.0	%	381	

Index	Parameter	Standard application	HVAC application	PID control application	Multi-pump (single drive) application	Multi-pump (multi-drive) application	Unit	ID	Description
3.5.2.1.6	AI1 Signal Inversion	0	0	0	0	0		387	
3.5.2.2.1	AI2 Signal Selection	101	101	101	101	101		388	101 = AnIN SlotA.2
3.5.2.2.2	AI2 Filter Time	0.1	0.1	0.1	0.1	0.1	s	389	
3.5.2.2.3	AI2 Signal Range	1	1	1	1	1		390	1 = 2...10 V/4...20 mA
3.5.2.2.4	AI2 Custom Min	0.0	0.0	0.0	0.0	0.0	%	391	
3.5.2.2.5	AI2 Custom Max	100.0	100.0	100.0	100.0	100.0	%	392	
3.5.2.2.6	AI2 Signal Inversion	0	0	0	0	0		398	
3.5.3.2.1	RO1 Function	2	2	2	49	2		11001	2 = Run
3.5.3.2.4	RO2 Function	3	3	3	50	3		11004	3 = Fault
3.5.3.2.7	RO3 Function	1	1	1	51	1		11007	1 = Ready
3.5.4.1.1	AO1 Function	2	2	2	2	2		10050	2 = Output Freq
3.5.4.1.2	AO1 Filter Time	1.0	1.0	1.0	1.0	1.0	s	10051	
3.5.4.1.3	AO1 Min Signal	0	0	0	0	0		10052	
3.5.4.1.4	AO1 Min Scale	0.0	0.0	0.0	0.0	0.0		10053	
3.5.4.1.5	AO1 Max Scale	0.0	0.0	0.0	0.0	0.0		10054	

Table 135: Default Values of Parameters of Group 3.10

Index	Parameter	Standard application	HVAC application	PID control application	Multi-pump (single drive) application	Multi-pump (multidrive) application	Unit	ID	Description
P3.10.1	Automatic Reset	0	0	1	1	1		731	0 = Disabled 1 = Enabled

Table 136: Default Values of Parameters of Group 3.13

Index	Parameter	Standard application	HVAC application	PID control application	Multi-pump (single drive) application	Multi-pump (multi-drive) application	Unit	ID	Description
P3.13.2.5	PID Setpoint Selection	0	0	0	102			1047	0 = DigIN Slot0.1

Index	Parameter	Standard application	HVAC application	PID control application	Multi-pump (single drive) application	Multi-pump (multi-drive) application	Unit	ID	Description
									102 = DigIN SlotA.3
3.13.2.6	PID Setpoint Source 1	-	-	1	1	1		332	1 = Keypad Setpoint 1
P3.13.2.10	PID Setpoint Source 2	-	-	-	-	2		431	2 = Keypad Setpoint 2
3.13.3.1	PID Feedback Function	-	-	1	1	1		333	1 = Only Source 1 in use
3.13.3.3	PID Feedback Source	-	-	2	2	2		334	2 = AI2

Table 137: Default Values of Parameters of Group 3.15

Index	Parameter	Standard application	HVAC application	PID control application	Multi-pump (single drive) application	Multi-pump (multi-drive) application	Unit	ID	Description
P3.15.1	Multi-pump Mode	-	-	-	0	2		1785	0= Single drive 1= Multifollower 2= Multimaster
P3.15.2	Number of Pumps	1	1	1	3	3		1001	
P3.15.5	Pump Interlocking	-	-	-	1	1		1032	0 = Disabled 1 = Enabled
P3.15.6	Autochange	-	-	-	1	1		1027	0 = Disabled 1 = Enabled (interval) 2 = Enabled (weekdays)
P3.15.7	Autochanged Pumps	-	-	-	1	1		1028	0 = Auxiliary Pump 1 = All Pumps
P3.15.8	Autochange Interval	-	-	-	48.0	48.0	h	1029	
P3.15.11	Autochange Frequency Limit	-	-	-	25.0	25.0	Hz	1031	
P3.15.12	Autochange Pump Limit	-	-	-	1	3		1030	
P3.15.13	Bandwidth	-	-	-	10.0	10.0	%	1097	

Index	Parameter	Standard application	HVAC application	PID control application	Multi-pump (single drive) application	Multi-pump (multi-drive) application	Unit	ID	Description
P3.15.14	Bandwidth Delay	-	-	-	10	10	s	1098	
P3.15.15	Constant Production Speed	-	-	-	-	100.0	%	1513	
P3.15.16	Running Pumps Limit	-	-	-	3	3		1187	

Table 138: Default Values of Parameters of Group 5.7

Index	Parameter	Standard application	HVAC application	PID control application	Multi-pump (single drive) application	Multi-pump (multidrive) application	Unit	ID	Description
P5.7.1	Timeout time	5	5	5	5	5	min	804	
P5.7.2	Default Page	4	4	4	4	4		2318	4 = Multi-monitor

12 Fault Tracing

12.1 Fault Types

When the control diagnostics of the AC drive find an unusual condition in the operation of the drive, the drive shows a notification about it. The notification can be seen on the display of the control panel. The display shows the code, the name, and a short description of the fault or alarm.

The source info tells the source of the fault, what caused it, where it occurred, and other data.

There are 3 different types of notification.

- An **info** does not affect the operation of the drive. The info must be reset.
- An **alarm** informs of unusual operation on the drive. It does not stop the drive. The alarm must be reset.
- A **fault** stops the drive. The drive must be reset and find a solution to the problem.

It is possible to program different responses for some faults in the application. See more in [6.9 Group 3.9: Protections](#).

Reset the fault with the [BACK/RESET] button on the keypad, or through the I/O terminal, fieldbus, or PC tool. The faults stay in the Fault history where they can be examined. See the different fault codes in [12.3 Fault Codes](#).

Before contacting the distributor or the factory because of unusual operation, prepare some data. Write down all the texts on the display, the fault code, the fault ID, the source info, the *Active Faults* list, and the *Fault History*.

12.2 Alarms and Warnings

In condition-based monitoring, for each feature, the user can define activation stages for warnings and alarms. The interpretation of alarm and warning color codes are as follows:

Green: No alarms are indicated. Condition-based monitoring operations continue.

Yellow: First indication of warning-stage 1 alarm is visible. Stage 1 warning fault are also shown as *Stage 1 Warning (S1)*. Notification to users to plan for maintenance operations. In this stage, condition-based monitoring operations continue.

Orange: Clear indication of warning-stage 2 alarm is visible. Stage 2 warning fault are also shown as *Stage 2 Warning (S2)*. Notification to users to act as soon as possible before the fault becomes critical.

Red: A critical alarm has occurred and condition-based monitoring operations has stopped.

12.3 Fault Codes

12.3.1 Fault Code 1, ID 1 - Overcurrent (Hardware Fault)

Cause

There is too high a current ($> 4 \times I_H$) in the motor cable.

- a sudden heavy load increase
- a short circuit in the motor cables
- the motor is not the correct type
- the parameter settings are not properly made

Troubleshooting

- Check the loading.
- Check the motor.
- Check the cables and connections.
- Make an identification run.
- Set the acceleration time longer (P3.4.1.2 and P3.4.2.2).

12.3.2 Fault Code 1, ID 2 - Overcurrent (Software Fault)

Cause

There is too high a current ($> 4 \times I_H$) in the motor cable.

- a sudden heavy load increase
- a short circuit in the motor cables
- the motor is not the correct type
- the parameter settings are not properly made

Troubleshooting

- Check the loading.
- Check the motor.
- Check the cables and connections.
- Make an identification run.
- Set the acceleration time longer (P3.4.1.2 and P3.4.2.2).

12.3.3 Fault Code 2, ID 10 - Overvoltage (Hardware Fault)

Cause

The DC link voltage is higher than the limits.

- too short a deceleration time
- high overvoltage spikes in the supply

Troubleshooting

- Set the deceleration time longer (P3.4.1.3 and P3.4.2.3).
- Activate the overvoltage controller.
- Check the input voltage.

12.3.4 Fault Code 2, ID 11 - Overvoltage (Software Fault)

Cause

The DC link voltage is higher than the limits.

- too short a deceleration time
- high overvoltage spikes in the supply

Troubleshooting

- Set the deceleration time longer (P3.4.1.3 and P3.4.2.3).
- Activate the overvoltage controller.
- Check the input voltage.

12.3.5 Fault Code 3, ID 20 - Earth Fault (Hardware Fault)

Cause

The measurement of current tells that the sum of the motor phase current is not zero.

- an insulation malfunction in the cables or the motor
- a filter (dU/dt, sine) malfunction

Troubleshooting

- Check the motor cables and the motor.
- Check the filters.

12.3.6 Fault Code 3, ID 21 - Earth fault (Software Fault)

Cause

The measurement of current tells that the sum of the motor phase current is not zero.

- an insulation malfunction in the cables or the motor
- a filter (dU/dt, sine) malfunction

Troubleshooting

- Check the motor cables and the motor.
- Check the filters.

12.3.7 Fault Code 5, ID 40 - Charging Switch

Cause

The charging switch is closed and the feedback information is OPEN.

- operation malfunction
- defective component

Troubleshooting

- Reset the fault and restart the drive.
- Check the feedback signal and the cable connection between the control board and the power board.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.8 Fault Code 7, ID 60 - Saturation

Cause

- Defective IGBT.
- De-saturation short circuit in the IGBT.
- A short circuit or an overload in the brake resistor.

Troubleshooting

- This fault cannot be reset from the control panel.
- Switch off the power.
- DO NOT RESTART THE DRIVE or CONNECT THE POWER!
- Ask for instructions from the factory.

12.3.9 Fault Code 8, ID 600 - System Fault

Cause

- There is no communication between the control board and the power unit.

Troubleshooting

- Reset the fault and restart the drive.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.10 Fault Code 8, ID 601 - System Fault

Cause

- There is no communication between the control board and the power.

Troubleshooting

- Reset the fault and restart the drive.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.11 Fault Code 8, ID 602 - System Fault

Cause

- Defective component.
- Operation malfunction.

Troubleshooting

- Reset the fault and restart the drive.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.12 Fault Code 8, ID 603 - System Fault

Cause

- Defective component.
- Operation malfunction.
- The voltage of auxiliary power in the power unit is too low.

Troubleshooting

- Reset the fault and restart the drive.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.13 Fault Code 8, ID 604 - System Fault

Cause

- Defective component.
- Operation malfunction.
- Output phase voltage does not agree to the reference.
- Feedback fault.

Troubleshooting

- Reset the fault and restart the drive.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.14 Fault Code 8, ID 605 - System Fault

Cause

- Defective component.
- Operation malfunction.

Troubleshooting

- Reset the fault and restart the drive.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.15 Fault Code 8, ID 606 - System Fault

Cause

- The software of the control unit is not compatible with the software of the power unit.

Troubleshooting

- Reset the fault and restart the drive.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.16 Fault Code 8, ID 607 - System Fault

Cause

- The software version cannot be read. There is no software in the power unit.
- Defective component.
- Operation malfunction (a problem in the power board or the measurement board).

Troubleshooting

- Reset the fault and restart the drive.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.17 Fault Code 8, ID 608 - System Fault

Cause

- A CPU overload.

Troubleshooting

- Reset the fault and restart the drive.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.18 Fault Code 8, ID 609 - System Fault

Cause

- Defective component.
- Operation malfunction.

Troubleshooting

- Reset the fault and make a power down of the drive twice.
- Download the latest software from the Danfoss Drives website. Update the drive with it.

12.3.19 Fault Code 8, ID 610 - System Fault

Cause

- Defective component.
- Operation malfunction.

Troubleshooting

- Reset the fault and restart.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.20 Fault Code 8, ID 614 - System Fault

Cause

- Configuration error.
- Software error.
- Defective component (a defective control board).
- Operation malfunction.

Troubleshooting

- Reset the fault and restart.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.21 Fault Code 8, ID 647 - System Fault

Cause

- Defective component.
- Operation malfunction.

Troubleshooting

- Reset the fault and restart.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.22 Fault Code 8, ID 648 - System Fault

Cause

- Operation malfunction.
- The system software is not compatible with the application.

Troubleshooting

- Reset the fault and restart.
- Download the latest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.23 Fault Code 8, ID 649 - System Fault

Cause

- A resource overload.
- A parameter loading, restoring or saving malfunction.

Troubleshooting

- Load the factory default settings.
- Download the latest software from the Danfoss Drives website. Update the drive with it.

12.3.24 Fault Code 8, ID 667 - System Fault

Cause

- Ethernet PHY is not recognized or it is in the wrong state.

Troubleshooting

- Reset the fault and restart the AC drive.
- Download the newest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.25 Fault Code 8, ID 670 - System Fault

Cause

- The output voltage is too low because of overload, a defective component, or a shortcut.

Troubleshooting

- Check the loading of the auxiliary output.
- Reset the fault and restart the AC drive.
- Download the newest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.26 Fault Code 8, ID 827 - System Fault

Cause

- Invalid/incorrect license key provided (via keypad or VCX).
- The license key is incorrect or not for this drive.

Troubleshooting

- Reset the fault and restart the AC drive.
- Enter the license key to the AC drive again.
- Download the newest software from the Danfoss Drives website. Update the drive with it.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.27 Fault Code 8, ID 828 - System Fault

Cause

- The entered license key was accepted and stored to the drive.

Troubleshooting

-

12.3.28 Fault Code 8, ID 829 - System Fault

Cause

- New licenses have been taken into use since the previous start-up.

Troubleshooting

-

12.3.29 Fault Code 8, ID 830 - System Fault

Cause

- Licenses have been removed from the drive.

Troubleshooting

-

12.3.30 Fault Code 9, ID 80 - Undervoltage (Fault)

Cause

The DC link voltage is lower than the limits.

- too low a supply voltage
- defective component
- a defective input fuse
- the external charge switch is not closed

NOTE! This fault becomes active only if the drive is in Run state.

Troubleshooting

- If there is a temporary supply voltage break, reset the fault and restart the drive.
- Do a check of the supply voltage. If the supply voltage is sufficient, there is an internal fault.
- Examine the electrical network for fault.
- Ask for instructions from your nearest distributor.

12.3.31 Fault Code 10, ID 91 - Input Phase

Cause

- supply voltage malfunction
- a defective fuse or malfunction in the supply cables

The load must be a minimum of 10–20% for the supervision to work.

Troubleshooting

- Check the supply voltage.
- Check the fuses.
- Check the supply cable.
- Check the rectifying bridge.
- Check the gate control of the thyristor (MR6->).

12.3.32 Fault Code 10, ID 92 - Input Phase

Cause

- Software-based input phase supervision.

Troubleshooting

- Check the supply voltage.
- Check the fuses.
- Check the supply cable.

NOTICE

This fault is only used with enclosure sizes MR9B, MR10, MR11, and MR12.

12.3.33 Fault Code 10, ID 94 - Input Phase**Cause**

- Software-based input phase supervision.

Troubleshooting

- Check the supply voltage.
- Check the fuses.
- Check the supply cable.

NOTICE

This fault is only used with enclosure sizes MR11 and MR12.

12.3.34 Fault Code 11, ID 100 - Output Phase Supervision**Cause**

The current measurement tells that there is no current in 1 motor phase.

- A motor or motor cables malfunction.
- Filter (dU/dt, sine wave) malfunction.

Troubleshooting

- Check the motor cable and the motor.
- Check the dU/dt or sine wave filter.

12.3.35 Fault Code 13, ID 120 - AC Drive Undertemperature (Fault)**Cause**

- Too low a temperature in the heat sink of the power unit or in the power board.

Troubleshooting

- The ambient temperature is too low for the drive. Move the drive in a warmer position.

12.3.36 Fault Code 14, ID 130 - AC Drive Overtemperature (Fault, Heat Sink)**Cause**

- Too high a temperature in the heat sink of the power unit or in the power board.
- The temperature limits of the heat sink are different in all the enclosure sizes.

Troubleshooting

- Check the actual amount and flow of cooling air.
- Examine the heat sink for dust.
- Check the ambient temperature.
- Make sure that the switching frequency is not too high in relation to the ambient temperature and the motor load.
- Check the cooling fan.

12.3.37 Fault Code 14, ID 131 - AC Drive Overtemperature (Alarm, Heat Sink)**Cause**

- Too high a temperature in the heat sink of the power unit or in the power board.
- The temperature limits of the heat sink are different in all the enclosure sizes.

Troubleshooting

- Check the actual amount and flow of cooling air.
- Examine the heat sink for dust.

- Check the ambient temperature.
- Make sure that the switching frequency is not too high in relation to the ambient temperature and the motor load.
- Check the cooling fan.

12.3.38 Fault Code 14, ID 132 - AC Drive Overtemperature (Fault, Board)

Cause

- Too high a temperature in the heat sink of the power unit or in the power board.
- The temperature limits of the heat sink are different in all the enclosure sizes.

Troubleshooting

- Check the actual amount and flow of cooling air.
- Examine the heat sink for dust.
- Check the ambient temperature.
- Make sure that the switching frequency is not too high in relation to the ambient temperature and the motor load.
- Check the cooling fan.

12.3.39 Fault Code 14, ID 133 - AC Drive Overtemperature (Alarm, Board)

Cause

- Too high a temperature in the heat sink of the power unit or in the power board.
- The temperature limits of the heat sink are different in all the enclosure sizes.

Troubleshooting

- Check the actual amount and flow of cooling air.
- Examine the heat sink for dust.
- Check the ambient temperature.
- Make sure that the switching frequency is not too high in relation to the ambient temperature and the motor load.
- Check the cooling fan.

12.3.40 Fault Code 14, ID 134 - AC Drive Overtemperature (Fault)

Cause

- Power unit auxiliary temperature is too high.

Troubleshooting

- Check the actual amount and flow of cooling air.
- Examine the heat sink for dust.
- Check the ambient temperature.
- Make sure that the switching frequency is not too high in relation to the ambient temperature and the motor load.
- Check the cooling fan.

12.3.41 Fault Code 14, ID 136 - Overvoltage Protection Circuit Temperature (Alarm)

Cause

- Too high an output capacitance or a ground fault in the floating network.

Troubleshooting

- Check the cables and the motor.

12.3.42 Fault Code 14, ID 137 - Overvoltage Protection Circuit Temperature (Fault)

Cause

- Too high an output capacitance or a ground fault in the floating network.

Troubleshooting

- Check the cables and the motor.

12.3.43 Fault Code 14, ID 138 - AC Drive Overtemperature (Fault)

Cause

- Output filter temperature is too high.

Troubleshooting

- Check the motor cables and the motor.
- Check the correct amount and flow of cooling air.
- Check the ambient temperature.

12.3.44 Fault Code 14, ID 139 - AC Drive Overtemperature (Fault)

Cause

- Brake chopper temperature is too high.

Troubleshooting

- Check the correct amount and flow of cooling air.
- Check the heat sink of the brake chopper for dust.
- Check the ambient temperature.

12.3.45 Fault Code 15, ID 140 - Motor Stall

Cause

- The motor stalled.

Troubleshooting

- Check the motor and the load.

12.3.46 Fault Code 16, ID 150 - Motor Overtemperature

Cause

- There is too heavy a load on the motor.

Troubleshooting

- Decrease the motor load. If there is no motor overload, check the motor thermal protection parameters (parameter group 3.9 Protections).

12.3.47 Fault Code 17, ID 160 - Motor Underload

Cause

- There is not a sufficient load on the motor.

Troubleshooting

- Check the load.
- Check the parameters.
- Check the dU/dt and sine wave filters.

12.3.48 Fault Code 18, ID 170 - Unbalance

Cause

- The current is unbalanced between power units.

Troubleshooting

- Check the motor cables and connections.
- Check the output filters (dU/dt filter, sine filter).
- Reset the fault and restart the drive.
- If the fault occurs again, ask for instructions from your nearest service center or distributor. Report carefully all the used software, application and options.

N O T I C E

This fault is only used with enclosure sizes MR11 and MR12.

12.3.49 Fault Code 18, ID 171 - Unbalance**Cause**

- The DC voltage is unbalanced between power units.

Troubleshooting

- Check the DC link cables between the power units.
- Check the motor cables and connections.
- Check the output filters (dU/dt filter, sine filter).
- Reset the fault and restart the drive.
- If the fault occurs again, ask for instructions from your nearest service center or distributor. Report carefully all the used software, application, and options.

N O T I C E

This fault is only used with enclosure sizes MR11 and MR12.

12.3.50 Fault Code 18, ID 173 - Unbalance (Alarm)**Cause**

- The current is unbalanced between power units.

Troubleshooting

- Check the installation and the motor cables.
- Redo the IGBT ID run.
- The event response can be set to No Action.

N O T I C E

This fault is only used with enclosure sizes MR11 and MR12.

12.3.51 Fault Code 19, ID 180 - Power Overload (Short-time Supervision)**Cause**

- The power of the drive is too high.

Troubleshooting

- Decrease the load.
- Examine the dimensions of drive. Examine if it is too small for the load.

12.3.52 Fault Code 19, ID 181 - Power Overload (Long-time Supervision)**Cause**

- The power of the drive is too high.

Troubleshooting

- Decrease the load.
- Examine the dimensions of drive. Examine if it is too small for the load.

12.3.53 Fault Code 25, ID 240 - Motor Control Fault**Cause**

This fault is available only if you use a customer-specific application. A malfunction in the start angle identification.

- The rotor moves during identification.
- The new angle does not agree with the old value.

Troubleshooting

- Reset the fault and restart the drive.
- Increase the identification current.
- See the fault history source for more information.

12.3.54 Fault Code 25, ID 241 - Motor Control Fault

Cause

This fault is available only if you use a customer-specific application. A malfunction in the start angle identification.

- The rotor moves during identification.
- The new angle does not agree with the old value.

Troubleshooting

- Reset the fault and restart the drive.
- Increase the identification current.
- See the fault history source for more information.

12.3.55 Fault Code 26, ID 250 - Start-up Prevented

Cause

- It is not possible to do a start-up of the drive. When the Run request is ON, a new software (a firmware or an application), a parameter setting, or other file that affects the operation of the drive, is loaded to drive.

Troubleshooting

- Reset the fault and stop the drive.
- Load the software and start the drive.

12.3.56 Fault Code 29, ID 280 - ATEX Thermistor

Cause

- The ATEX thermistor tells that there is an overtemperature.

Troubleshooting

- Reset the fault. Check the thermistor and its connections.

12.3.57 Fault Code 30, ID 290 - Safe Off

Cause

- The safe Off signal A does not let you set the drive to the READY state.

Troubleshooting

- Reset the fault and restart the drive.
- Check the signals from the control board to the power unit and the D connector.

12.3.58 Fault Code 30, ID 291 - Safe Off

Cause

- The safe Off signal B does not let you set the drive to the READY state.

Troubleshooting

- Reset the fault and restart the drive.
- Check the signals from the control board to the power unit and the D connector.

12.3.59 Fault Code 30, ID 500 - Safety Configuration

Cause

- The safety configuration switch was installed.

Troubleshooting

- Remove the safety configuration switch from the control board.

12.3.60 Fault Code 30, ID 501 - Safety Configuration

Cause

- There are too many STO option boards. It is possible to have only 1.

Troubleshooting

- Keep 1 of the STO option boards. Remove the others. See the safety manual.

12.3.61 Fault Code 30, ID 502 - Safety Configuration

Cause

- The STO option board was installed in an incorrect slot.

Troubleshooting

- Put the STO option board into the correct slot. See the OPTBJ STO Safety manual.

12.3.62 Fault Code 30, ID 503 - Safety configuration

Cause

- There is no safety configuration switch on the control board.

Troubleshooting

- Install the safety configuration switch on the control board. See the safety manual.

12.3.63 Fault Code 30, ID 504 - Safety configuration

Cause

- The safety configuration switch was installed incorrectly on the control board.

Troubleshooting

- Install the safety configuration switch into the correct position on the control board. See the safety manual.

12.3.64 Fault Code 30, ID 505 - Safety configuration

Cause

- The safety configuration switch was installed incorrectly on the STO option board.

Troubleshooting

- Check the installation of the safety configuration switch on the STO option board. See the safety manual.

12.3.65 Fault Code 30, ID 506 - Safety configuration

Cause

- There is no communication with the STO option board.

Troubleshooting

- Check the installation of the STO option board. See the safety manual.

12.3.66 Fault Code 30, ID 507 - Safety configuration

Cause

- The STO option board is not compatible with the hardware.

Troubleshooting

- Reset the drive and restart it. If the fault occurs again, ask for instructions from your nearest distributor.

12.3.67 Fault Code 30, ID 520 - Safety diagnostics

Cause

- The STO inputs have a different status.

Troubleshooting

- Check the external safety switch. Check the input connection and cable of the safety switch.
- Reset the drive and restart.
- If the fault occurs again, ask for instructions from your nearest distributor.

12.3.68 Fault Code 30, ID 521 - Safety diagnostics

Cause

- A malfunction in the ATEX thermistor diagnostic. There is no connection in the ATEX thermistor input.

Troubleshooting

- Reset the drive and restart.
- If the fault occurs again, change the option board.

12.3.69 Fault Code 30, ID 522 - Safety diagnostics

Cause

- A short circuit in the connection of the ATEX thermistor input.

Troubleshooting

- Check the ATEX thermistor input connection.
- Check the external ATEX connection.
- Check the external ATEX thermistor.

12.3.70 Fault Code 30, ID 530 - Safe Torque Off

Cause

- An emergency stop was connected or some other STO operation was activated.

Troubleshooting

- When the STO function is activated, the drive is in safe state.

12.3.71 Fault Code 32, ID 311 - Fan Cooling

Cause

- The fan speed does not agree to the speed reference accurately, but the drive operates correctly.
- This fault shows only in the MR7 and in enclosure sizes bigger than MR7.

Troubleshooting

- Reset the fault and restart the drive.
- Clean or replace the fan.

12.3.72 Fault Code 32, ID 312 - Fan Cooling

Cause

- The fan life time (that is, 50 000 h) is complete.

Troubleshooting

- Replace the fan and reset the life time counter of the fan.

12.3.73 Fault Code 33, ID 320 - Fire Mode Enabled

Cause

- The fire mode of the drive is enabled. The protections of the drive are disabled or autoreset.

Troubleshooting

- Check the parameter settings and the signals.
- This alarm is reset automatically when fire mode is disabled.

12.3.74 Fault Code 37, ID 361 - Device Changed (Same Type)

Cause

- The power unit was replaced by a new one that has the same size. The device is ready to be used. The parameters are available in the drive.

Troubleshooting

- Reset the fault. The drive reboots after you reset the fault.

12.3.75 Fault Code 37, ID 362 - Device Changed (Same Type)

Cause

- The option board in slot B was replaced by a new one that you have used before in the same slot. The device is ready to be used.

Troubleshooting

- Reset the fault. The drive starts to use the old parameter settings.

12.3.76 Fault Code 37, ID 363 - Device Changed (Same Type)

Cause

- The option board in slot C was replaced by a new one that you have used before in the same slot. The device is ready to be used.

Troubleshooting

- Reset the fault. The drive starts to use the old parameter settings.

12.3.77 Fault Code 37, ID 364 - Device Changed (Same Type)

Cause

- The option board in slot D was replaced by a new one that you have used before in the same slot. The device is ready to be used.

Troubleshooting

- Reset the fault. The drive starts to use the old parameter settings.

12.3.78 Fault Code 37, ID 365 - Device Changed (Same Type)

Cause

- The option board in slot E was replaced by a new one that you have used before in the same slot. The device is ready to be used.

Troubleshooting

- Reset the fault. The drive starts to use the old parameter settings.

12.3.79 Fault Code 38, ID 372 - Device Added (Same Type)

Cause

- An option board was put into slot B. You have used the option board before in the same slot.

Troubleshooting

- The device is ready for use. The drive starts to use the old parameter settings.

12.3.80 Fault Code 38, ID 373 - Device Added (Same Type)

Cause

- An option board was put into slot C. You have used the option board before in the same slot.

Troubleshooting

- The device is ready for use. The drive starts to use the old parameter settings.

12.3.81 Fault Code 38, ID 374 - Device Added (Same Type)

Cause

- An option board was put into slot D. You have used the option board before in the same slot.

Troubleshooting

- The device is ready for use. The drive starts to use the old parameter settings.

12.3.82 Fault Code 38, ID 375 - Device Added (Same Type)

Cause

- An option board was put into slot E. You have used the option board before in the same slot.

Troubleshooting

- The device is ready for use. The drive starts to use the old parameter settings.

12.3.83 Fault Code 39, ID 382 - Device Removed

Cause

- An option board was removed from slot A or B.

Troubleshooting

- The device is not available. Reset the fault.

12.3.84 Fault Code 39, ID 383 - Device Removed

Cause

- An option board was removed from slot C.

Troubleshooting

- The device is not available. Reset the fault.

12.3.85 Fault Code 39, ID 384 - Device Removed

Cause

- An option board was removed from slot D.

Troubleshooting

- The device is not available. Reset the fault.

12.3.86 Fault Code 39, ID 385 - Device Removed

Cause

- An option board was removed from slot E.

Troubleshooting

- The device is not available. Reset the fault.

12.3.87 Fault Code 40, ID 390 - Device Unknown

Cause

- An unknown device was connected (the power unit/option board).

Troubleshooting

- The device is not available. If the fault occurs again, ask for instructions from your nearest distributor.

12.3.88 Fault Code 41, ID 400 - IGBT Temperature

Cause

The calculated IGBT temperature is too high.

- too high a motor load
- too high an ambient temperature
- hardware malfunction

Troubleshooting

- Check the parameter settings.
- Examine the actual amount and flow of cooling air.
- Check the ambient temperature.

- Examine the heatsink for dust.
- Make sure that the switching frequency is not too high in relation to the ambient temperature and the motor load.
- Check the cooling fan.
- Make an identification run.

12.3.89 Fault Code 44, ID 431 - Device Changed (Different Type)

Cause

- There is a new power unit of a different type. Parameters are not available in the settings.

Troubleshooting

- Reset the fault. The drive reboots after you reset the fault.
- Set the power unit parameters again.

12.3.90 Fault Code 44, ID 433 - Device Changed (Different Type)

Cause

- The option board in slot C was replaced by a new one that you have not used before in the same slot. No parameter settings are saved.

Troubleshooting

- Reset the fault. Set the option board parameters again.

12.3.91 Fault Code 44, ID 434 - Device Changed (Different Type)

Cause

- The option board in slot D was replaced by a new one that you have not used before in the same slot. No parameter settings are saved.

Troubleshooting

- Reset the fault. Set the option board parameters again.

12.3.92 Fault Code 44, ID 435 - Device Changed (Different Type)

Cause

- The option board in slot E was replaced by a new one that you have not used before in the same slot. No parameter settings are saved.

Troubleshooting

- Reset the fault. Set the option board parameters again.

12.3.93 Fault Code 45, ID 441 - Device Added (Different Type)

Cause

- There is a new power unit of a different type. Parameters are not available in the settings.

Troubleshooting

- Reset the fault. The drive reboots after you reset the fault.
- Set the power unit parameters again.

12.3.94 Fault Code 45, ID 443 - Device Added (Different Type)

Cause

- A new option board, that you have not used before in the same slot, was put in slot C. No parameter settings are saved.

Troubleshooting

- Set the option board parameters again.

12.3.95 Fault Code 45, ID 444 - Device Added (Different Type)

Cause

- A new option board, that you have not used before in the same slot, was put in slot D. No parameter settings are saved.

Troubleshooting

- Set the option board parameters again.

12.3.96 Fault Code 45, ID 445 - Device Added (Different Type)

Cause

- A new option board, that you have not used before in the same slot, was put in slot E. No parameter settings are saved.

Troubleshooting

- Set the option board parameters again.

12.3.97 Fault Code 46, ID 662 - Real Time Clock

Cause

- The voltage of the RTC battery is low.

Troubleshooting

- Replace the battery.

12.3.98 Fault Code 47, ID 663 - Software Updated

Cause

- The software of the drive was updated, the full software package or an application.

Troubleshooting

- No steps are necessary.

12.3.99 Fault Code 50, ID 1050 - AI Low Fault

Cause

- 1 or more of the available analog input signals is below 50% of the minimum signal range.
- A control cable is defective or loose.
- A malfunction in a signal source.

Troubleshooting

- Replace the defective parts.
- Check the analog input circuit.
- Make sure that parameter AI1 Signal Range is set correctly.

12.3.100 Fault Code 51, ID 1051 - Device External Fault

Cause

- The digital input signal that is set with parameter P3.5.1.11 or P3.5.1.12 was activated.

Troubleshooting

- This is a user-defined fault.
- Check the digital inputs and schematics.

12.3.101 Fault Code 52, ID 1052 - Keypad Communication Fault

Cause

- The connection between the control panel and the drive is defective.

Troubleshooting

- Check the control panel connection and the control panel cable if you have it.

12.3.102 Fault Code 52, ID 1352 - Keypad Communication Fault

Cause

- The connection between the control panel and the drive is defective.

Troubleshooting

- Check the control panel connection and the control panel cable if you have it.

12.3.103 Fault Code 53, ID 1053 - Fieldbus Communication Fault

Cause

- The data connection between the fieldbus master and the fieldbus board is defective.

Troubleshooting

- Check the installation and fieldbus master.

12.3.104 Fault Code 53, ID 1054 - Redundant Switchover

Cause

- The data connection between controlling fieldbus and the drive has timed out. Redundant control has changed the controlling fieldbus connection.

Troubleshooting

- Check the installation and the communication cables.

12.3.105 Fault Code 53, ID 1055 - Redundant Timeout

Cause

- The data connection between controlling fieldbus protocols and the drive has timed out.

Troubleshooting

- Check the installation and the communication cables.

12.3.106 Fault Code 54, ID 1354 - Slot A fault

Cause

- A defective option board or slot.

Troubleshooting

- Check the board and the slot.
- Ask for instructions from your nearest distributor.

12.3.107 Fault Code 54, ID 1454 - Slot B fault

Cause

- A defective option board or slot.

Troubleshooting

- Check the board and the slot.
- Ask for instructions from your nearest distributor.

12.3.108 Fault Code 54, ID 1554 - Slot C fault

Cause

- A defective option board or slot.

Troubleshooting

- Check the board and the slot.
- Ask for instructions from your nearest distributor.

12.3.109 Fault Code 54, ID 1654 - Slot D fault

Cause

- A defective option board or slot.

Troubleshooting

- Check the board and the slot.
- Ask for instructions from your nearest distributor.

12.3.110 Fault Code 54, ID 1754 - Slot E fault

Cause

- A defective option board or slot.

Troubleshooting

- Check the board and the slot.
- Ask for instructions from your nearest distributor.

12.3.111 Fault Code 57, ID 1057 - Identification

Cause

- There was a failure in the identification run.

Troubleshooting

- Make sure that the motor is connected to the drive.
- Make sure that there is no load on the motor shaft.
- Make sure that the start command is not removed before the identification run is complete.

12.3.112 Fault Code 57, ID 1157 - Identification

Cause

- During the identification run, drive was not able to reach required frequency reference.

Troubleshooting

- Make sure that minimum and maximum frequency references are set correctly. Too low maximum frequency can prevent drive from reaching required frequency.

12.3.113 Fault Code 57, ID 1257 - Identification

Cause

- During the identification run, drive was not able to reach required frequency reference.

Troubleshooting

- Make sure that acceleration time is set correctly. Too long acceleration time can prevent drive from reaching required frequency in 40 seconds.

12.3.114 Fault Code 57, ID 1357 - Identification

Cause

- During the identification run, drive was not able to reach required frequency reference.

Troubleshooting

- Make sure that current, torque, and power limits of the drive are set correctly. Too low limit settings can prevent drive from reaching required frequency.

12.3.115 Fault Code 63, ID 1063 - Quick Stop Fault

Cause

- The quick stop function is activated.

Troubleshooting

- Find the cause for the quick stop activation. After you find it, correct it. Reset the fault and restart the drive.
- See parameter P3.5.1.26 and the quick stop parameters.

12.3.116 Fault Code 63, ID 1363 - Quick Stop Alarm

Cause

- The quick stop function is activated.

Troubleshooting

- Find the cause for the quick stop activation. After you find it, correct it. Reset the fault and restart the drive.
- See parameter P3.5.1.26 and the quick stop parameters.

12.3.117 Fault Code 65, ID 1065 - PC Communication Fault

Cause

- The data connection between the PC and the drive is defective.

Troubleshooting

- Check the installation, cable, and terminals between the PC and the drive.

12.3.118 Fault Code 66, ID 1366 - Thermistor Input 1 Fault

Cause

- The motor temperature increased.

Troubleshooting

- Check the motor cooling and the load.
- Check the thermistor connection.
- If the thermistor input is not used, you have to short-circuit it.
- Ask for instructions from your nearest distributor.

12.3.119 Fault Code 66, ID 1466 - Thermistor Input 2 Fault

Cause

- The motor temperature increased.

Troubleshooting

- Check the motor cooling and the load.
- Check the thermistor connection.
- If the thermistor input is not used, you have to short-circuit it.
- Ask for instructions from your nearest distributor.

12.3.120 Fault Code 66, ID 1566 - Thermistor Input 3 Fault

Cause

- The motor temperature increased.

Troubleshooting

- Check the motor cooling and the load.
- Check the thermistor connection.
- If the thermistor input is not used, you have to short-circuit it.
- Ask for instructions from your nearest distributor.

12.3.121 Fault Code 68, ID 1301 - Maintenance Counter 1 Alarm

Cause

- The value of the maintenance counter is higher than the alarm limit.

Troubleshooting

- Do the necessary maintenance.
- Reset the counter. See parameter B3.16.4 or P3.5.1.40.

12.3.122 Fault Code 68, ID 1302 - Maintenance Counter 1 Fault

Cause

- The value of the maintenance counter is higher than the fault limit.

Troubleshooting

- Do the necessary maintenance.
- Reset the counter. See parameter B3.16.4 or P3.5.1.40.

12.3.123 Fault Code 68, ID 1303 - Maintenance Counter 2 Alarm

Cause

- The value of the maintenance counter is higher than the alarm limit.

Troubleshooting

- Do the necessary maintenance.
- Reset the counter. See parameter B3.16.4 or P3.5.1.40.

12.3.124 Fault Code 68, ID 1304 - Maintenance Counter 2 Fault

Cause

- The value of the maintenance counter is higher than the fault limit.

Troubleshooting

- Do the necessary maintenance.
- Reset the counter. See parameter B3.16.4 or P3.5.1.40.

12.3.125 Fault Code 69, ID 1310 - Fieldbus Mapping Error

Cause

- The ID number that is used to map the values to Fieldbus Process Data Out is not valid.

Troubleshooting

- Check the parameters in the Fieldbus Data menu.

12.3.126 Fault Code 69, ID 1311 - Fieldbus Mapping Error

Cause

- It is not possible to convert 1 or more values for Fieldbus Process Data Out.
- The type of the value is undefined.

Troubleshooting

- Check the parameters in the Fieldbus Data menu.

12.3.127 Fault Code 69, ID 1312 - Fieldbus Mapping Error

Cause

- There is an overflow when the values for Fieldbus Process Data Out (16-bit) are mapped and converted.

Troubleshooting

- Check the parameters in the Fieldbus Data menu.

12.3.128 Fault Code 70, ID 1120 - BaselineRun Failed

Cause

- There was a failure during the baseline run.

Troubleshooting

- Do a check of the parameters in the baseline run settings.

12.3.129 Fault Code 71, ID 1121 - Stator Winding: Current Warning S1

Cause

- Current unbalance has exceeded the detection high limit. Early fault detected in motor.

Troubleshooting

- Check motor stator windings.

12.3.130 Fault Code 71, ID 1122 - Stator Winding: Current Warning S2

Cause

- Current unbalance has exceeded the detection high limit. Severe fault can soon occur in the motor.

Troubleshooting

- Check motor stator windings.

12.3.131 Fault Code 71, ID 1123 - Stator Winding: Current Alarm/Fault

Cause

- Current unbalance has exceeded the detection high limit. Severe fault is detected in motor.

Troubleshooting

- Check motor stator windings.

12.3.132 Fault Code 71, ID 1127 - Vibration Warning S1

Cause

- Vibration has exceeded the detection high limit. Early fault detected in motor.

Troubleshooting

- Check root cause for increased vibration.

12.3.133 Fault Code 71, ID 1128 - Vibration Warning S2

Cause

- Vibration has exceeded the detection high limit. Severe fault can occur soon in the motor.

Troubleshooting

- Check root cause for increased vibration.

12.3.134 Fault Code 71, ID 1129 - Vibration Alarm/Fault

Cause

- Vibration has exceeded the detection high limit. Severe fault detected in motor.

Troubleshooting

- Check root cause for increased vibration.

12.3.135 Fault Code 71, ID 1130 - Load Warning S1

Cause

- Load outside of the window limits. Early fault detected in motor.

Troubleshooting

- Check root cause for excessive overload or underload.

12.3.136 Fault Code 71, ID 1131 - Load Warning S2

Cause

- Load outside of the window limits. Severe fault can occur soon in the motor.

Troubleshooting

- Check root cause for excessive overload or underload.

12.3.137 Fault Code 71, ID 1132 - Load Alarm/Fault

Cause

- Load outside of the window limits. Severe fault is detected in motor.

Troubleshooting

- Check root cause for excessive overload or underload.

12.3.138 Fault Code 76, ID 1076 - Start Prevented

Cause

- The start command is blocked to prevent the accidental rotation of the motor during the first power-up.

Troubleshooting

- Reset the drive to start the correct operation. The parameter settings tell if it is necessary to restart the drive.

12.3.139 Fault Code 77, ID 1077 - > 5 Connections

Cause

- There are more than 5 active fieldbus or PC tool connections. You can use only 5 connections at the same time.

Troubleshooting

- Leave 5 active connections. Remove the other connections.

12.3.140 Fault Code 78, ID 1078 - Identification Ongoing

Cause

- The drive is running the identification for the motor.

Troubleshooting

- You can stop the identification run at any time by removing the start command. When the identification run is completed, a new start command is needed to start the drive.

12.3.141 Fault Code 100, ID 1100 - Soft Fill Timeout

Cause

- There is a timeout in the Soft fill function in the PID controller. The process value was not achieved in the time limit.
- A pipe that broke can be the cause.

Troubleshooting

- Check the process.
- Check the parameters in the menu M3.13.8.

12.3.142 Fault Code 101, ID 1101 - Feedback Supervision Fault (PID 1)

Cause

- The feedback value is not in the supervision limits (P3.13.6.2 and P3.13.6.3) and the delay (P3.13.6.4), if you set the delay.

Troubleshooting

- Check the process.
- Check the parameter settings, the supervision limits, and the delay.

12.3.143 Fault Code 105, ID 1105 - Feedback Supervision Fault (ExtPID)

Cause

- The feedback value is not in the supervision limits (P3.14.4.2 and P3.14.4.3) and the delay (P3.14.4.4), if you set the delay.

Troubleshooting

- Check the process.
- Check the parameter settings, the supervision limits, and the delay.

12.3.144 Fault Code 109, ID 1109 - Input Pressure Supervision

Cause

- The supervision signal of the input pressure (P3.13.9.2) is lower than the alarm limit (P3.13.9.7).

Troubleshooting

- Check the process.
- Check the parameters in menu M3.13.9.
- Check the input pressure sensor and connections.

12.3.145 Fault Code 109, ID 1409 - Input Pressure Supervision

Cause

- The supervision signal of the input pressure (P3.13.9.2) is lower than the fault limit (P3.13.9.8).

Troubleshooting

- Check the process.
- Check the parameters in menu M3.13.9.
- Check the input pressure sensor and connections.

12.3.146 Fault Code 111, ID 1315 - Temperature Fault 1

Cause

- 1 or more of the temperature input signals (set in P3.9.6.1) is higher than the alarm limit (P3.9.6.2).

Troubleshooting

- Find the cause of the temperature rise.
- Check the temperature sensor and connections.
- If no sensor is connected, make sure that the temperature input is hardwired.
- See the option board manual for more data.

12.3.147 Fault Code 111, ID 1316 - Temperature Fault 1

Cause

- 1 or more of the temperature input signals (set in P3.9.6.1) is higher than the fault limit (P3.9.6.3).

Troubleshooting

- Find the cause of the temperature rise.
- Check the temperature sensor and connections.
- If no sensor is connected, make sure that the temperature input is hardwired.
- See the option board manual for more data.

12.3.148 Fault Code 112, ID 1317 - Temperature Fault 2

Cause

- 1 or more of the temperature input signals (set in P3.9.6.5) is higher than the fault limit (P3.9.6.6).

Troubleshooting

- Find the cause of the temperature rise.
- Check the temperature sensor and connections.
- If no sensor is connected, make sure that the temperature input is hardwired.
- See the option board manual for more data.

12.3.149 Fault Code 112, ID 1318 - Temperature Fault 2

Cause

- 1 or more of the temperature input signals (set in P3.9.6.5) is higher than the fault limit (P3.9.6.7).

Troubleshooting

- Find the cause of the temperature rise.
- Check the temperature sensor and connections.
- If no sensor is connected, make sure that the temperature input is hardwired.
- See the option board manual for more data.

12.3.150 Fault Code 113, ID 1113 - Pump Running Time

Cause

- In the Multi-pump system, 1 or more of the pump run-time counters is above a user-specified alarm limit.

Troubleshooting

- Do the necessary maintenance actions, reset the run-time counter, and reset the alarm. See Pump running time counters.

12.3.151 Fault Code 113, ID 1313 - Pump Running Time

Cause

- In the Multi-pump system, 1 or more of the pump run-time counters is above a user-specified alarm limit.

Troubleshooting

- Do the necessary maintenance actions, reset the run-time counter, and reset the alarm. See Pump running time counters.

12.3.152 Fault Code 117, ID 1117 - Flow Switch

Cause

- The feedback signal from the flow switch was inactive longer than the specified flow switch fault delay time.

Troubleshooting

- Do a check of the process.
- Do a check of the parameter settings.

12.3.153 Fault Code 118, ID 1118 - AHF Over Temp

Cause

- The advanced harmonic filter function has caused an overtemperature fault through a digital input.

Troubleshooting

- Check the advanced harmonic filter function.

12.3.154 Fault Code 119, ID 1119 - Auto Cleaning

Cause

- The motor did not reach the expected speed reference during the auto cleaning sequence.

Troubleshooting

- Check the process.
- Check the auto cleaning parameter settings.

12.3.155 Fault Code 300, ID 700 - Unsupported

Cause

- The application is not compatible (it is unsupported).

Troubleshooting

- Replace the application.

12.3.156 Fault Code 300, ID 701 - Unsupported

Cause

- The option board or the slot is not compatible (it is unsupported).

Troubleshooting

- Remove the option board.

12.4 Total and Trip Counters

The AC drive has different counters based on the operation time of the drive and the energy consumption. Some of the counters measure total values and some can be reset.

The energy counters measure the energy that is taken from the supply network. The other counters are used to measure, for example, the operation time of the drive or the run-time of the motor.

It is possible to monitor all the counter values from the PC, control panel, or fieldbus. When using the PC or the control panel, it is possible to monitor the counter values in the Diagnostics menu. When using fieldbus, it is possible to read the counter values with the ID numbers.

12.4.1 Operating Time Counter

Resetting: not possible

Location: submenu Total counters

The value of the counter has 5 different 16-bit values. To read the value of the counter through fieldbus, use these ID numbers.

ID number	Counter value
ID 1754	Operating Time Counter (years)
ID 1755	Operating Time Counter (days)
ID 1756	Operating Time Counter (hours)
ID 1757	Operating Time Counter (minutes)
ID 1758	Operating Time Counter (seconds)

Example

The value 1a 143d 02:21 is received from the operating time counter from the fieldbus.

- ID1754: 1 (years)
- ID1755: 143 (days)
- ID1756: 2 (hours)
- ID1757: 21 (minutes)
- ID1758: 0 (seconds)

12.4.2 Operating Time Trip Counter

The operating time trip counter of the control unit can be reset. It is in the submenu Trip counters. It is possible to reset the counter with the PC, the control panel, or the fieldbus. The value of the counter has 5 different 16-bit values. To read the value of the counter through fieldbus, use these ID numbers.

ID number	Counter value
ID 1766	Operating Time Trip Counter (years)
ID 1767	Operating Time Trip Counter (days)
ID 1768	Operating Time Trip Counter (hours)
ID 1769	Operating Time Trip Counter (minutes)
ID 1770	Operating Time Trip Counter (seconds)

Example

The value 1a 143d 02:21 is received from the operating time trip counter from the fieldbus.

- ID1766: 1 (years)
- ID1767: 143 (days)
- ID1768: 2 (hours)
- ID1769: 21 (minutes)
- ID1770: 0 (seconds)

ID number	Counter value
ID 2311	Operating Time Trip Counter Reset

To reset the energy trip counter, use the PC, the control panel, or the fieldbus. When using the PC or the control panel, reset the counter in the Diagnostics menu. When using the fieldbus, set a rising edge to ID 2311 Operating Time Trip Counter Reset.

12.4.3 Run Time Counter

The run time counter of the motor cannot be reset. It is in the submenu Total counters. The value of the counter has 5 different 16-bit values. To read the value of the counter through fieldbus, use these ID numbers.

ID number	Counter value
ID 1772	Run Time Counter (years)
ID 1773	Run Time Counter (days)
ID 1774	Run Time Counter (hours)
ID 1775	Run Time Counter (minutes)
ID 1776	Run Time Counter (seconds)

Example

The value 1a 143d 02:21 is received from the run time counter from the fieldbus.

- ID1772: 1 (years)
- ID1773: 143 (days)
- ID1774: 2 (hours)
- ID1775: 21 (minutes)
- ID1776: 0 (seconds)

12.4.4 Power on Time Counter

The power on time counter of the power unit is in the submenu Total counters. It is not possible to reset the counter. The value of the counter has 5 different 16-bit values. To read the value of the counter through fieldbus, use these ID numbers.

ID number	Counter value
ID 1777	Power On Time Counter (years)
ID 1778	Power On Time Counter (days)
ID 1779	Power On Time Counter (hours)
ID 1780	Power On Time Counter (minutes)
ID 1781	Power On Time Counter (seconds)

Example

The value 1a 240d 02:18 is received from the power on time counter from the fieldbus.

- ID1777: 1 (years)
- ID1778: 240 (days)
- ID1779: 2 (hours)
- ID1780: 18 (minutes)
- ID1781: 0 (seconds)

12.4.5 Energy Counter

The energy counter counts the total quantity of energy that the drive gets from the supply network. The counter cannot be reset. To read the value of the counter through fieldbus, use these ID numbers.

ID number	Counter value
ID 2291	Energy Counter

The value always has 4 digits. The format and the unit of the counter change to agree with the energy counter value.

Example

- 0.001 kWh
- 0.010 kWh

- 0.100 kWh
- 1.000 kWh
- 10.00 kWh
- 100.0 kWh
- 1.000 MWh
- 10.00 MWh
- 100.0 MWh
- 1.000 GWh
- and so on

ID number	Counter value
ID 2303	Energy Counter Format

The energy counter format gives the position of the decimal point in the value of the Energy Counter.

- 40 = 4 digits, 0 fractional digits
- 41 = 4 digits, 1 fractional digit
- 42 = 4 digits, 2 fractional digits
- 43 = 4 digits, 3 fractional digits

Example

- 0.001 kWh (Format = 43)
- 100.0 kWh (Format = 41)
- 10.00 MWh (Format = 42)

ID number	Counter value
ID 2305	Energy Counter Unit

The energy counter unit gives the unit for the value of the Energy Counter.

- 0 = kWh
- 1 = MWh
- 2 = GWh
- 3 = TWh
- 4 = PWh

Example

If the value 4500 is received from ID 2291, the value 42 from ID 2303, and the value 0 from ID 2305, the result is 45.00 kWh.

12.4.6 Energy Trip Counter

The energy trip counter counts the quantity of energy that the drive gets from the supply network. The counter is in the submenu Trip counters. It is possible to reset the counter with the PC, the control panel, or the fieldbus. To read the value of the counter through fieldbus, use these ID numbers.

ID number	Counter value
ID 2296	Energy Trip Counter

The value has always 4 digits. The format and the unit of the counter change to agree with the energy trip counter value. See the following example. It is possible to monitor the energy counter format and unit with ID 2307 Energy Trip Counter Format and ID 2309 Energy trip Counter unit.